DPUTimeDistance

This DPU computes the distance in seconds to another vehicle.

1. Inputs:

LongDist	Longitudinal distance to the vehicle
SimCarSpeed	Speed of the EGO vehicle
LongSpeed	Speed of the other vehicle

2. Static parameters:

ValueMissing	Return value for ‚no vehicle'. Default: -9999
ValueSimCarSpeedLow	Return value for ‚EGO vehicle too slow'. Default: -8888
DistOfs	Offset to the measured value. Default: 3.44

3. Outputs:

TimeDistance	Distance in seconds to vehicle
TTC	Time-to-Collision to vehicle

4. Functional principle:

If speed *SimCarSpeed* > 0.01 m/s, output *TimeDistance* is computed with following formula:

$$TimeDistance = \frac{LongDist - DistOfs}{SimCarSpeed}$$

If a *LongDist* of 0.0m is input, value *ValueMissing* is returned.

If speed (Input *SimCarSpeed*) is less than 0.01 m/s, value *ValueSimCarSpeedLow* is returned.

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